

Video Transcript

Video 3: Basics of a CDC System (02:41)

Let's build a change data capture system that is a CDC system. We're going to write or we're going to pretend that we have a very simple application that simply writes a timestamp into the database. As you can see here, it will be written into MySQL. We're then going to react by detecting the change, capturing the change, and then propagating it throughout the tiers that you see there. MySQL will propagate to MongoDB, that will be propagated to Redis, and that will be propagated to Cassandra. We will use a periodic queries approach and you can see here the flow chart. We will start off by doing a query. If we detect the change, then we will select the new data, and then we will propagate that data that is save or share. When it comes to the organization, MySQL will be the system of record, the source of truth, and then we will be repeating that the tag capture and propagate through the tiers. Now, as you can imagine, writing something like this could become quite involved. You are creating containers, you are checking on the logic. There's a lot that could be going on here when it comes to creating that entire system to be able to track those changes.

So, because of it, we're going to write an automated system, and we're going to break it down into components to make it more granular, to be able to test, and to be able to ensure that we have, in fact, the functionality that we think we do. So, we're going to write all of this in Python, and there will be a number of components. The first one will be one through which on which through Python, we will be writing shell commands. We will be able to create containers through that as well and then delete the containers. When we're moving through a fast development cycle, we want to be able to create and tear that down. That is, clean up what we are creating while we are creating our CDC system. Just as well, some databases require you to initialize them before they can be populated, so we will do that programmatically as well. Then we will clean up the databases. In some cases, we don't want to delete the entire container. We don't want to delete the database or the table. We simply want to clear out the data. Finally, we will do all of this in an event loop. That is a timer that will be going through all of these steps, and once again we will control all of this using Python.